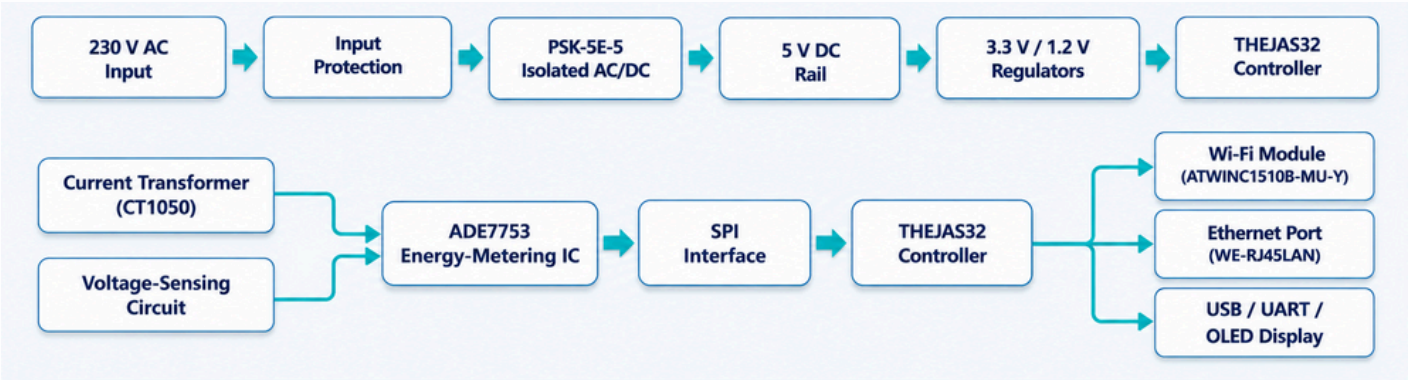
Single-Phase Energy Monitoring IoT Gateway

Problem statement :

Design a PCB-based Energy Monitoring IoT Gateway capable of acquiring electrical parameters from an electrical system, processing the collected data, and communicating the information to an external monitoring or control system.

The board should be designed as a complete hardware platform that integrates electrical measurement interfaces, signal conditioning, processing, power management, communication, protection, and debugging capabilities required for energy monitoring.

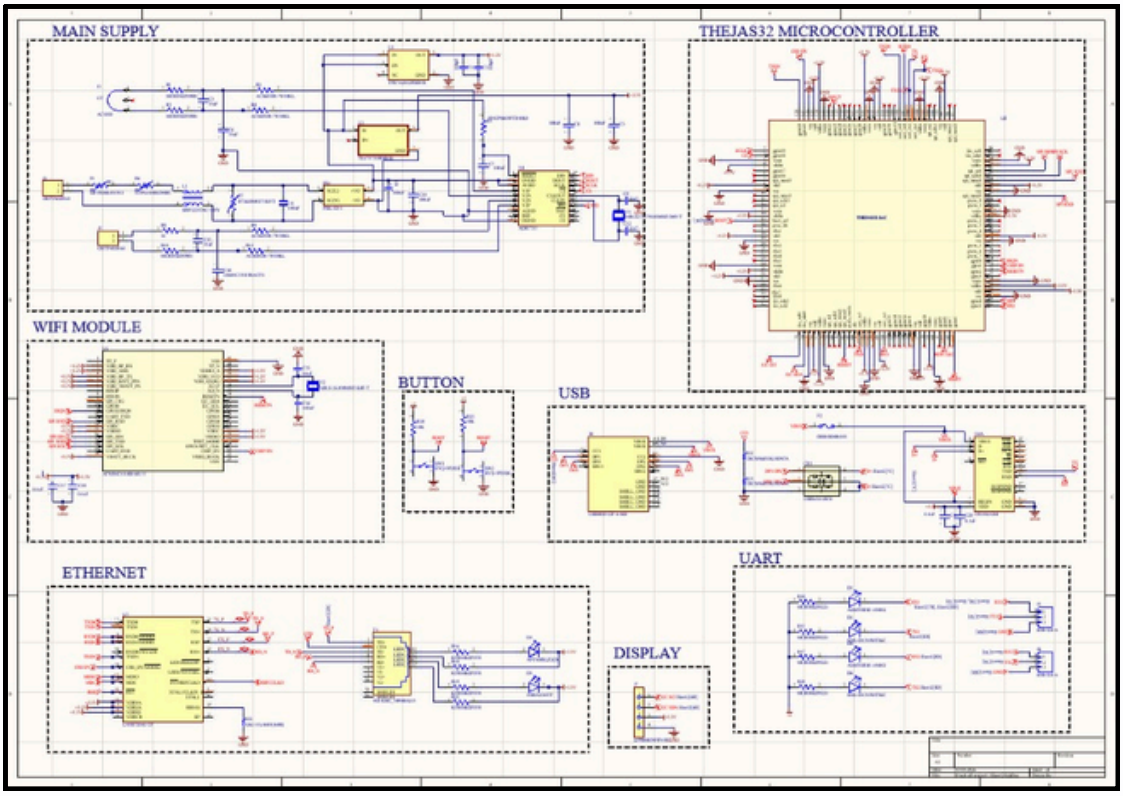
Proposed problem solution :



Components :

S. No.	Component name	Part number	Package
1	Energy-metering IC	ADE7753	SSOP-20
2	Microcontroller	THEJAS32	LQFP-128
3	Current transformer	CT1050	THT
4	Isolated AC/DC power module	PSK-5E-5	THT
5	Ethernet port	WE-RJ45LAN	Through-hole PCB mount
6	wifi module	ATWINC1510B-MU-Y	QFN-32

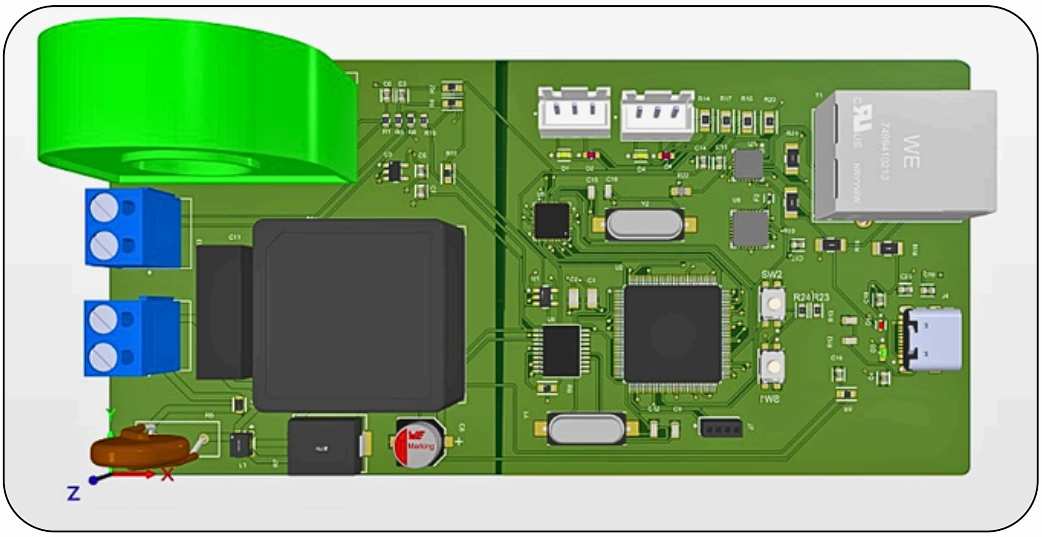
schematic diagram



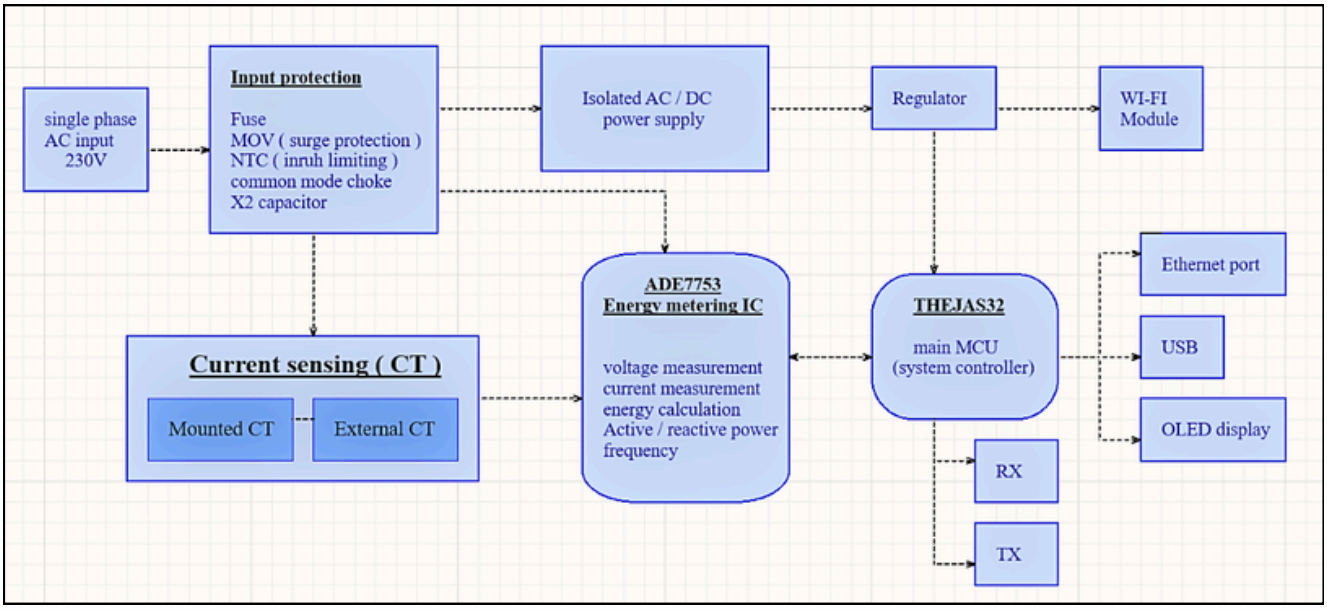
BOARD SIZE :

- Total Distance: 136.878 mm ÷ 10
  = 13.6878 cm
- X Distance: 123.635 mm ÷ 10
  = 12.3635 cm
- Y Distance: 58.737 mm ÷ 10
  = 5.8737 cm

3D VIEW



Block Diagram



General operating conditions (THEJAS32)

SYMBOL	PARAMETER	MIN	NOR	MAX
vdd	core voltage	1.08	1.2	1.32
vddio	supply voltage	2.97	3.3	3.36
temp	operating temperature range	0	+25	+70

Application

- **Industrial energy monitoring:** Monitor power consumption of machines and equipment.
- **Smart energy management:** Track energy usage and identify high-consumption periods.
- **Remote monitoring:** Send measurements to a monitoring application through Wi-Fi or Ethernet, if implemented.
- **Energy consumption analysis:** Record historical data and analyze usage patterns.
- **Electrical load monitoring:** Observe changes in current and power during equipment operation.
- **Building energy monitoring:** Monitor single-phase electrical loads in buildings and facilities.